## ■ 4.2.2 BASIC

One similarity between BASIC and *Mathematica* is that both are primarily interpreted languages. You type input, and they immediately process it.

*Mathematica* is, of course, a much more sophisticated language than BASIC. However, at a simple level, you can use *Mathematica* a bit like a version of BASIC that does symbolic, as well numerical, computations.

One immediate difference is that *Mathematica* prints the result from every input line you give. You do not have to include a notation like `?` to get your results printed out.

| BASIC | *Mathematica* |
|---|---|
| *f*(*x*) | *f*[*x*] |
| *x* * *y* | *x* * *y* or *x* *y* |
| `let` *x* = *y* (assignment) | *x* = *y* |
| *x* = *y* (equality test) | *x* == *y* |
| *x* <> *y* | *x* != *y* |
| *x* `and` *y* | *x* && *y* |
| *x* `or` *y* | *x* \|\| *y* |
| `sqr`(*x*) | `Sqrt`[*x*] |
| `fix`(*x*) | `Floor`[*x*] |
| `int`(*x*) | `If`[*x*>=0, `Floor`[*x*], `Ceiling`[*x*]] |
| `rnd`() | `Random`[ ] |
| `abs`(*x*), `log`(*x*) etc. | `Abs`[*x*], `Log`[*x*], etc. |
| *a*(*i*) (array reference) | *a*[[*i*]] |
| `rem` *text* | (* *text* *) |
| `if` *predicate* `then` *t* `else` *f* | `If`[*predicate*, *t*, *f*] |
| `for` *i* = *imin* `to` *imax* ↩ *statement* ↩ `next` *i* | `Do`[*statement*, {*i*, *imin*, *imax*}] |

Some correspondences between BASIC and *Mathematica*.

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